



PERFECT SOLUTIONS

Unleashing IT Talent

Functional Requirements Document

Internal HRMS Implementation

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Prepared By	IT & Product Implementation Team
Prepared For	Perfect Solutions

Version	Date	Author	Reviewer	Change
0.1	2026-01-18	Lead Engineer	Project Manager	Initial draft from BRD baseline
0.5	2026-01-26	Lead Engineer	PS Engineering Manager	Tech review
1.0	2026-05-12	Project Manager	PS Steering Committee	Approved baseline

1.1 Reference Documents

- PS-BRD-2026-001 — Perfect Solutions Business Requirements Document, v1.0
- Perfect Solutions HRMS Architecture Specification (Internal)
- Perfect Solutions HRMS Security & Compliance Whitepaper (Internal)
- Perfect Solutions Brand Guidelines (Internal)

2. Introduction

2.1 Purpose

This Functional Requirements Document (FRD) translates the business requirements captured in the BRD into discrete, testable functional specifications for the Perfect Solutions HRMS platform deployed for Perfect Solutions. Each requirement is uniquely identified, mapped back to one or more business requirements, and accompanied by acceptance criteria that form the basis of UAT.

2.2 Intended Audience

- Perfect Solutions IT and Engineering teams (technical reviewers)
- Perfect Solutions HR Operations (process validators)
- Internal delivery, engineering, and QA teams (implementers)
- Internal and external auditors (compliance reviewers)

2.3 Document Conventions

- Functional requirements prefixed FR-NNN; non-functional NFR-NNN.
- Use cases prefixed UC-NNN; user stories US-NNN.
- Priority follows MoSCoW.
- Each requirement traces back to a BRD identifier (BR-NNN).

3. System Overview

The Perfect Solutions HRMS is a multi-tenant SaaS platform that comprising a React single-page application, a FastAPI backend, and a MongoDB persistence tier on Kubernetes. For Perfect Solutions the platform is delivered as a fully white-labeled tenant identified by the slug 'perfectsolutions', accessible at hr.perfectsolutions.com.

Figure 1 — System Architecture (Multi-Tenant SaaS)

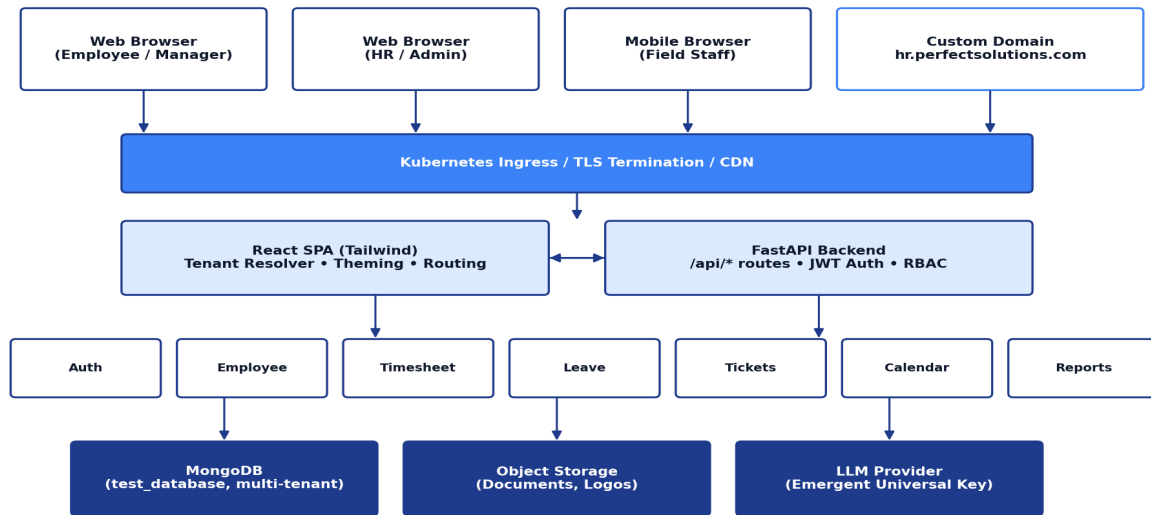


Figure 1 — System Architecture (Multi-Tenant SaaS)

3.1 Technology Stack

Layer	Technology	Purpose
Frontend	React 18, TailwindCSS, react-router-dom, Framer Motion	UI, animations, routing
State	React Context API, hooks	Auth, tenant, theming
Backend	FastAPI (Python 3.11), Pydantic v2	REST API, validation, OpenAPI
Database	MongoDB (test_database)	Tenant-scoped persistence
Auth	JWT (HS256), bcrypt	Stateless authentication
AI	Emergent Universal Key (Claude / GPT / Gemini)	AI assistant inside HRMS
PDF	ReportLab	Server-side document generation
Calendar	react-big-calendar	Calendar UX
Hosting	Kubernetes (Emergent Platform)	Orchestration, ingress, TLS

3.2 Tenant Resolution & White-Labeling

- Tenant resolution via slug path or custom domain (hr.perfectsolutions.com).
- Frontend calls GET /api/tenants/by-slug/{slug} or /by-domain to fetch metadata.
- Tenant metadata drives logo, colors, browser tab title, and feature flags.
- All API calls implicitly carry tenant context via JWT 'tenant_id' claim.

4. User Roles & Permission Matrix

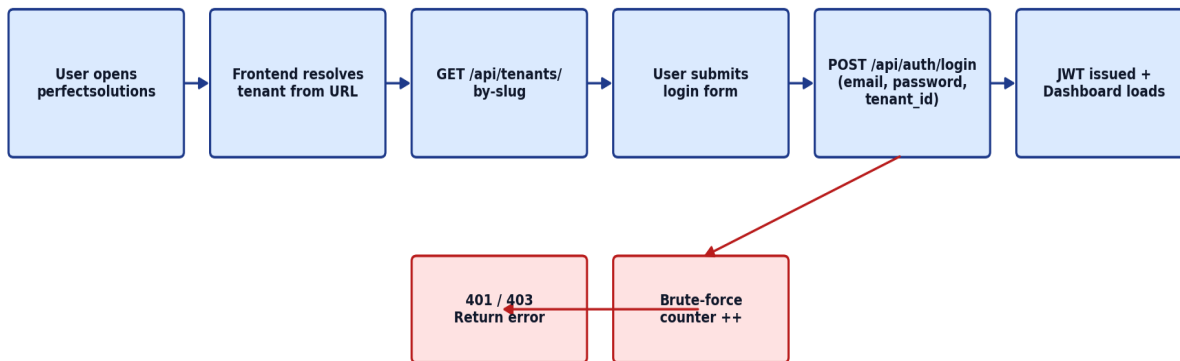
Roles below are scoped to the Perfect Solutions tenant. Super Admin is platform-level and reserved for IT operations.

Capability	Super Admin	Tenant Admin	HR Mgr.	People Mgr.	Employee
Manage tenants & branding	✓	—	—	—	—
Create/Edit Employees	—	✓	✓	—	—
Approve Timesheet	—	✓	✓	✓ (team)	—
Approve Leave	—	✓	✓	✓ (team)	—
Submit Timesheet / Leave	—	✓	✓	✓	✓
Raise Tickets	—	✓	✓	✓	✓
Resolve Tickets	—	✓	✓	✓ (assigned)	—
View All Reports	✓	✓	✓	Team only	Self only
Configure Departments / Leave Types	—	✓	✓	—	—
Audit Log Access	✓	✓	Read-only	—	—

5. Functional Modules

5.1 Authentication & Onboarding

Figure 2 — Authentication & Tenant Resolution Flow



On invalid credentials → log attempt, increment fail counter, lock account after 5 attempts within 15 minutes.

Figure 2 — Authentication & Tenant Resolution Flow

FR ID	Functional Requirement	Pri.	Trace
FR-001	Allow new admin signup only with valid tenant-specific invite code.	Must	BR-002
FR-002	Validate code against tenants collection; reject with 'Invalid admin invite code'.	Must	BR-002
FR-003	Hash passwords using bcrypt with work factor ≥ 12.	Must	BR-019
FR-004	Issue JWT containing user_id, tenant_id, role, expiry on successful login.	Must	BR-003
FR-005	Lock account after 5 failed attempts within 15 min; lock duration 30 min.	Must	BR-020
FR-006	Allow password reset via tokenized link sent to registered email.	Should	BR-019
FR-007	Expire idle sessions after 30 min of inactivity.	Must	BR-019
FR-008	Display Perfect Solutions branding on all auth pages.	Must	BR-001

5.1.1 Use Case — UC-001 Admin Self-Onboarding

Field	Detail
Actor	Prospective Tenant Admin at Perfect Solutions
Pre-condition	Valid admin invite code received via secure channel.
Trigger	User navigates to hr.perfectsolutions.com/signup
Main Flow	1) User enters name, email, password, admin code. 2) System validates code against perfectsolutions tenant. 3) System creates user with role TENANT_ADMIN. 4) JWT issued; user redirected to admin dashboard.
Alt Flow A1	Invite code invalid → display error and abort signup.
Alt Flow A2	Email already registered → prompt to log in instead.
Post-condition	User account created, audit log entry written.
Acceptance	Given a valid admin code, when the user submits the signup form with valid data, then the account is created and the dashboard loads within 2 seconds.

5.2 Employee Management

FR ID	Functional Requirement	Pri.	Trace
FR-010	HR creates an employee record with name, email, dept, role, manager, join date.	Must	BR-005
FR-011	HR edits any employee field except immutable identifiers.	Must	BR-005
FR-012	HR deactivates (soft-delete) an employee; deactivated users cannot log in.	Must	BR-005
FR-013	HR searches and filters employees by name, dept, manager, status.	Should	BR-005
FR-014	Employees view and update profile photo, contact, and emergency contact.	Must	BR-024
FR-015	Enforce email uniqueness within a tenant.	Must	BR-005
FR-016	Department list tenant-configurable; defaults: General, Engineering, HR, Sales, Marketing, Finance.	Must	BR-005

5.3 Time & Attendance

FR ID	Functional Requirement	Pri.	Trace
FR-020	Employees log time entries per day with project, task, hours.	Must	BR-006
FR-021	Employees save drafts and submit weekly.	Must	BR-006
FR-022	Managers approve, reject (with reason), or request changes.	Must	BR-007
FR-023	Flag timesheets > 60 hours/week as exceptions.	Should	BR-006
FR-024	Approved timesheets locked from edits except by HR.	Must	BR-007
FR-025	Weekly utilization report per project.	Should	BR-015
FR-026	Time entries exportable in CSV/Excel.	Must	BR-015

5.4 Leave Management

FR ID	Functional Requirement	Pri.	Trace
FR-030	Employees view current leave balance per leave type.	Must	BR-008
FR-031	Employees submit leave requests with type, start, end, reason.	Must	BR-008
FR-032	Reject requests exceeding balance or overlapping approved leave.	Must	BR-010
FR-033	Managers approve/reject with optional comments.	Must	BR-009
FR-034	Approved leave deducts balance and adds calendar entry visible to team.	Must	BR-013
FR-035	Email + in-app notifications on submit, approval, rejection.	Must	BR-017
FR-036	HR configures leave types, accrual, carry-over per leave type.	Must	BR-008
FR-037	Default leave types: Vacation, Sick, Personal, Unpaid.	Must	BR-008

Figure 3 — Leave Request Workflow

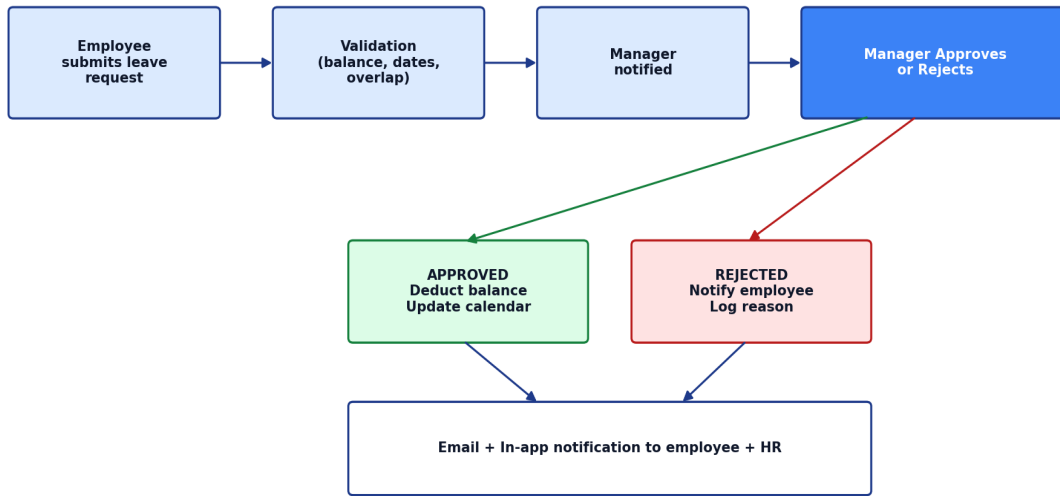


Figure 3 — Leave Workflow

5.5 Tickets / Helpdesk

FR ID	Functional Requirement	Pri.	Trace
FR-040	Employees raise tickets with subject, description, category, priority, attachments.	Must	BR-011
FR-041	Tickets auto-routed by category to dept queue.	Must	BR-011
FR-042	Agents comment, change status, reassign, close.	Must	BR-011
FR-043	Track first-response and resolution SLA, warn when at risk.	Should	BR-012
FR-044	Notify requesters on status change; reopen within 7 days.	Must	BR-017
FR-045	Threaded chat-style conversation per ticket (polling).	Must	BR-011

5.6 Calendar & Scheduling

FR ID	Functional Requirement	Pri.	Trace
FR-050	Unified calendar combining approved leaves, company events, personal events.	Must	BR-013
FR-051	Users create personal/team events with title, start, end, attendees.	Must	BR-013
FR-052	Managers view team availability heatmap by week.	Should	BR-016
FR-053	Calendar offers Day, Week, Month, Agenda views.	Must	BR-013

5.7 Document Vault

FR ID	Functional Requirement	Pri.	Trace
FR-060	Employees download personal documents (offer letters, payslips, policy ack).	Must	BR-014
FR-061	HR uploads company-wide policy documents available to all employees.	Must	BR-014
FR-062	Documents stored encrypted at rest, accessible only to authorized roles.	Must	BR-019
FR-063	Document downloads audit-logged.	Must	BR-018

5.8 Performance Reviews

FR ID	Functional Requirement	Pri.	Trace
FR-070	Managers create review cycles with self, peer, and manager assessments.	Should	BG-07
FR-071	Employees set quarterly goals (OKRs) and update progress.	Should	BG-07
FR-072	HR views aggregate performance trends by department.	Could	BG-10

5.9 Reports & Analytics

FR ID	Functional Requirement	Pri.	Trace
FR-080	Standard reports: Headcount, Attendance, Leave Utilization, Ticket SLA, Project Utilization.	Must	BR-015
FR-081	Filter by date range, department, project.	Must	BR-015
FR-082	Export reports as CSV and Excel.	Must	BR-015
FR-083	Dashboards refresh in near real-time (< 1 min lag).	Should	BR-016

6. User Stories with Acceptance Criteria

ID	User Story	Acceptance Criteria
US-01	As an Employee, I want to submit a leave request from my phone so that I do not need to be at my desk.	Given I am authenticated on a mobile browser, when I open Leave → New Request, then I can submit and see a confirmation toast within 2 seconds.
US-02	As a Manager, I want a single approval queue so that I do not switch contexts between leave, time, and tickets.	Given I have pending approvals across modules, when I open My Approvals, then I see a unified list ordered by SLA risk.
US-03	As HR, I want to bulk import employees from a spreadsheet so that I avoid manual entry during onboarding waves.	Given a CSV in the standard template, when I upload it, then valid rows are imported and errors are downloadable as a per-row report.
US-04	As an Admin, I want browser tab titles to display 'Perfect Solutions — HR Portal' so brand identity is preserved.	Given any page on hr.perfectsolutions.com, then the tab title contains 'Perfect Solutions' and never a generic platform string.
US-05	As an Auditor, I want to download an immutable audit log for any date range so that I can reconcile changes.	Given Auditor role, when I request the export, then a CSV with timestamp, actor, action, before/after states is produced.
US-06	As an Employee, I want to see my real-time leave balance before I submit a request so that I avoid rejections.	Given my leave page, when I select a leave type and dates, then the projected remaining balance is shown live.
US-07	As a Manager, I want to be notified within 1 minute of an SLA-at-risk ticket on my queue.	Given a ticket within 80% of its SLA window, when polling fires, then the agent receives an in-app warning and an email.
US-08	As HR, I want to configure leave types per policy so that I can adapt to state-specific sick-leave laws.	Given Tenant Admin role, when I edit leave types, then changes apply prospectively without affecting historical balances.

7. Data Model

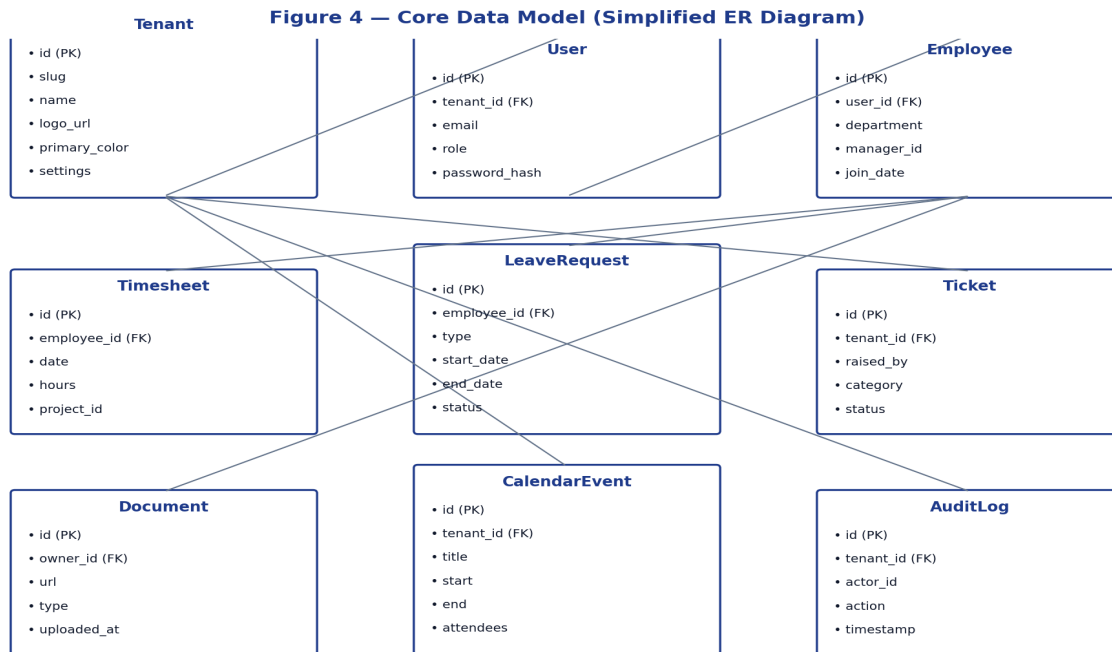


Figure 4 — Core Data Model (Simplified ERD)

All collections are tenant-scoped: every persisted document carries an indexed `tenant_id` field. Backend queries automatically constrain by JWT-resolved `tenant_id`; integration tests verify no API call may return another tenant's data.

7.1 Key Collections

Collection	Cardinality	Indexes	Notes
tenants	1 per company	slug (unique), custom_domain (unique sparse)	Branding, settings, invite code
users	1..N per tenant	tenant_id+email (unique)	Auth + role
employees	1..1 with users	tenant_id, manager_id	Profile data
timesheets	Many	tenant_id+employee_id+date	Locked after approval
leave_requests	Many	tenant_id+employee_id, status	Approval workflow
tickets	Many	tenant_id+status, assigned_to	SLA-tracked
calendar_events	Many	tenant_id+start	Includes leave-derived events
documents	Many	tenant_id+owner_id	Encrypted at rest
audit_logs	Many (append-only)	tenant_id+timestamp	Immutable

8. API Surface (Representative)

All backend endpoints are prefixed with /api and require a valid JWT (except the explicitly public ones). Tenant context is derived from the JWT's tenant_id claim. The full OpenAPI specification is generated by FastAPI at /api/docs.

Method	Endpoint	Purpose	Auth
POST	/api/auth/signup	Tenant admin sign-up with invite code	Public
POST	/api/auth/login	Login, returns JWT	Public
GET	/api/tenants/by-slug/{slug}	Resolve tenant by slug	Public
GET	/api/tenants/by-domain	Resolve tenant by custom domain	Public
GET	/api/employees	List tenant employees	Authenticated
POST	/api/employees	Create employee	Tenant Admin / HR
GET	/api/timesheets	List timesheets (filtered)	Authenticated
POST	/api/timesheets	Create / submit timesheet	Employee+
POST	/api/timesheets/{id}/approve	Approve timesheet	Manager+
GET	/api/leave/balance	Get leave balance	Authenticated
POST	/api/leave/request	Submit leave request	Employee+
POST	/api/leave/{id}/approve	Approve leave	Manager+
POST	/api/tickets	Create ticket	Authenticated
GET	/api/tickets/{id}	Get ticket detail incl. comments	Authenticated
GET	/api/calendar/events	List events for visible range	Authenticated
GET	/api/reports/{type}	Generate report (CSV/JSON)	HR / Admin
GET	/api/audit-logs	Read-only audit log	Admin / Auditor

9. Non-Functional Requirements

NFR ID	Category	Requirement	Target
NFR-001	Performance	API response under normal load	p95 < 500 ms; p99 < 1.2 s
NFR-002	Performance	Time-to-interactive on dashboard	< 2.5 s on 4G
NFR-003	Availability	Monthly uptime (excl. planned)	≥ 99.9%
NFR-004	Scalability	Concurrent active users / tenant	≥ 1,000
NFR-005	Scalability	Records per collection	≥ 10 million
NFR-006	Security	Password hashing	bcrypt (cost ≥ 12)
NFR-007	Security	Transport encryption	TLS 1.2+
NFR-008	Security	Storage encryption	AES-256 at rest for PII
NFR-009	Security	Tenant isolation	100% verified by tests
NFR-010	Privacy	Data subject access response	≤ 30 days
NFR-011	Reliability	RPO	≤ 1 hour
NFR-012	Reliability	RTO	≤ 4 hours
NFR-013	Usability	Accessibility	WCAG 2.1 AA
NFR-014	Compatibility	Supported browsers	Latest 2 versions
NFR-015	Mobile	Responsive layout	≥ 360 px width
NFR-016	Audit	Audit log retention	≥ 7 years
NFR-017	Maintainability	Backend test coverage	≥ 80%
NFR-018	Observability	Logs + metrics + traces	100% endpoints
NFR-019	Localization	Default time zone	America/New_York
NFR-020	Compliance	Audit-ready evidence collection	≤ 5 business days

10. Functional Risk Analysis

ID	Risk	Probability	Impact	Mitigation
FRK-01	Race condition on leave balance during concurrent submissions	Medium	High	Optimistic concurrency check; server-authoritative deduction in single transaction.
FRK-02	Browser tab title regressions on tenant context change	Medium	Low	Centralized useBrowserTitle hook keyed on tenant resolution.
FRK-03	Custom domain CNAME misconfiguration	Medium	Medium	Pre-flight DNS check + admin-visible verification status.
FRK-04	Incomplete role enforcement on new endpoints	Medium	High	RBAC decorator pattern enforced via lint + integration tests.
FRK-05	MongoDB ObjectId leakage in API responses	Low	Medium	Pydantic response models, _id excluded by default, contract tests.
FRK-06	Duplicate /auth/signup routes leading to inconsistent validation	Confirmed	Medium	Refactor server.py to remove duplicate; consolidate into routes/auth.py.
FRK-07	Email deliverability for notifications	Medium	Medium	Authenticated SMTP/Resend with SPF, DKIM, DMARC.

11. Traceability Matrix (BR → FR → UC / US)

Every business requirement is satisfied by one or more functional requirements, validated by use cases or user stories, and verified by acceptance test cases.

Business Req.	Functional Req.	Use Cases / Stories	Test Method
BR-001 Branding	FR-008	UC-001, US-04	Visual + automated regression
BR-002 Invite-only signup	FR-001, FR-002	UC-001	Functional + negative tests
BR-003 Auth required	FR-004, FR-007	UC-001	API + UI session tests
BR-004 RBAC	FR-001..FR-008, role matrix	UC-001	Role permutation tests
BR-005 Employee CRUD	FR-010..FR-016	US-03	CRUD test suite
BR-006/7 Time	FR-020..FR-026	US-02	Workflow + approval tests
BR-008..10 Leave	FR-030..FR-037	US-01, US-06, US-08	Workflow + balance tests
BR-011/12 Tickets	FR-040..FR-045	US-07	SLA + workflow tests
BR-013 Calendar	FR-050..FR-053	US-01	Cross-module integration
BR-014 Documents	FR-060..FR-063	—	Permission + audit tests
BR-015/16 Reports	FR-080..FR-083	—	Snapshot + export tests
BR-017 Notifications	FR-035, FR-044	US-07	Email + in-app tests
BR-018 Audit	FR-063 + audit_logs	US-05	Append-only verification
BR-019/20 Security	FR-003..FR-005, NFR-006..009	—	Pen test + automated checks
BR-021 Availability	NFR-003	—	Synthetic monitoring
BR-022 Performance	NFR-001, NFR-002	—	Load test
BR-023 Accessibility	NFR-013	—	Axe / Lighthouse audit
BR-024 Self-service	FR-014, FR-031, FR-040	US-01	End-to-end flow tests
BR-025 Integrations	API surface	—	Contract tests
BR-026 Privacy	NFR-010	US-05	DSAR drill
BR-027 Localization	NFR-019	—	TZ regression tests
BR-028 Mobile	NFR-015	US-01	Mobile viewport tests
BR-029 Tenant isolation	NFR-009	—	Cross-tenant fuzz tests
BR-030 DR	NFR-011, NFR-012	—	DR drill

12. UAT Strategy & Exit Criteria

- Three UAT cycles: smoke, full regression, and pre-prod soak.
- Test data set: 50 sample employees, 5 managers, 3 departments to start.
- Defect severities: S1 Blocker, S2 Major, S3 Minor, S4 Cosmetic.
- Exit criteria: Zero S1, zero S2, < 5 S3 open at sign-off.
- Performance gate: NFR-001/002 met under 2× expected load.
- Security gate: Pen test report with zero High findings outstanding.

13. Glossary

Term	Definition
FR / NFR	Functional / Non-Functional Requirement
JWT	JSON Web Token — signed, stateless auth token
MoSCoW	Must / Should / Could / Won't priority framework
RBAC	Role-Based Access Control
SLA	Service Level Agreement
UAT	User Acceptance Testing
WCAG	Web Content Accessibility Guidelines
RPO/RTO	Recovery Point / Time Objectives
DSAR	Data Subject Access Request

14. Approval & Sign-off

By signing below, the parties acknowledge that this Functional Requirements Document accurately translates the business needs into implementable functional and non-functional specifications.

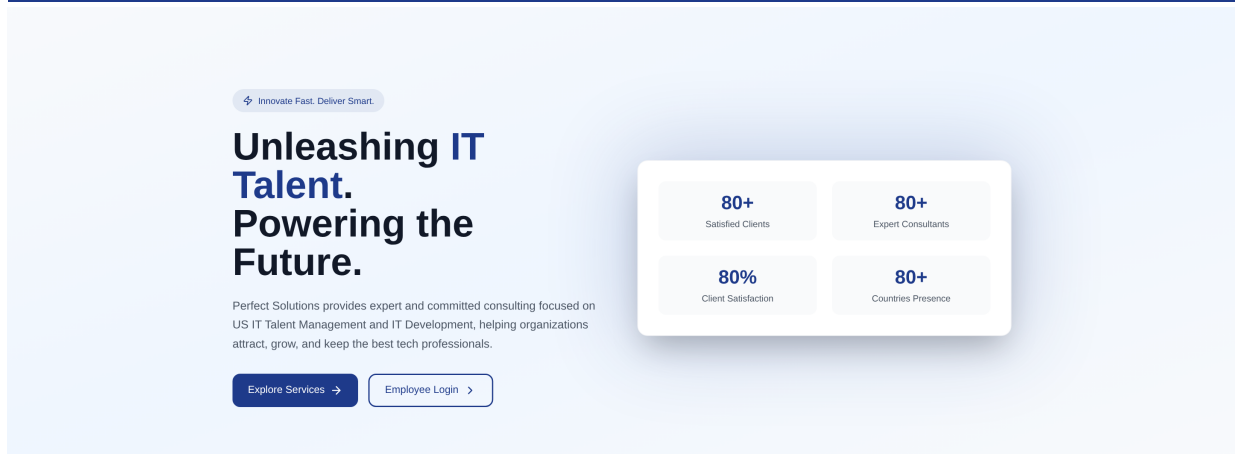
Role	Name	Signature	Date
Engineering Manager	Perfect Solutions	_____	_____
IT Security Officer	Perfect Solutions	_____	_____
Head of HR Operations	Perfect Solutions	_____	_____
Lead Engineer	Internal Engineering	_____	_____
Product Manager	Product Office	_____	_____

Appendix A — Portal Walkthrough (Screenshots)

This appendix presents annotated screen captures from the live Perfect Solutions tenant of the Perfect Solutions HRMS portal. Screens are grouped by role to illustrate how the platform satisfies the requirements stated earlier in this document. All screens are rendered on the **perfectsolutions** tenant with full brand styling (logo, primary color **#1E3A8A**) applied automatically.

A.1 Public / Tenant-Branded Pages

Screen A1 — Tenant Landing

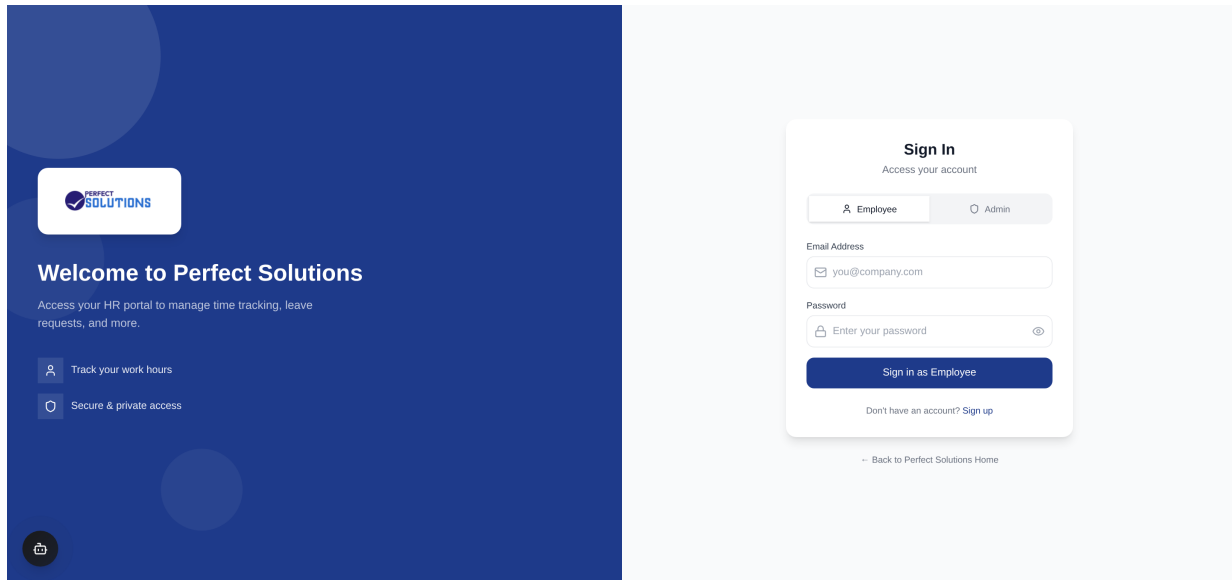


Professional & Dedicated Consulting Solutions

We combine deep industry knowledge with customized strategies across strategy, operations, and HR to deliver clear, powerful results and long-term success.

Branded landing page for hr.perfectsolutions.com — logo, brand colors, and tagline applied automatically by the multi-tenant theming engine.

Screen A2 — Login



Welcome to Perfect Solutions

Access your HR portal to manage time tracking, leave requests, and more.

- Track your work hours
- Secure & private access

Sign In
Access your account

Employee Admin

Email Address
you@company.com

Password
Enter your password

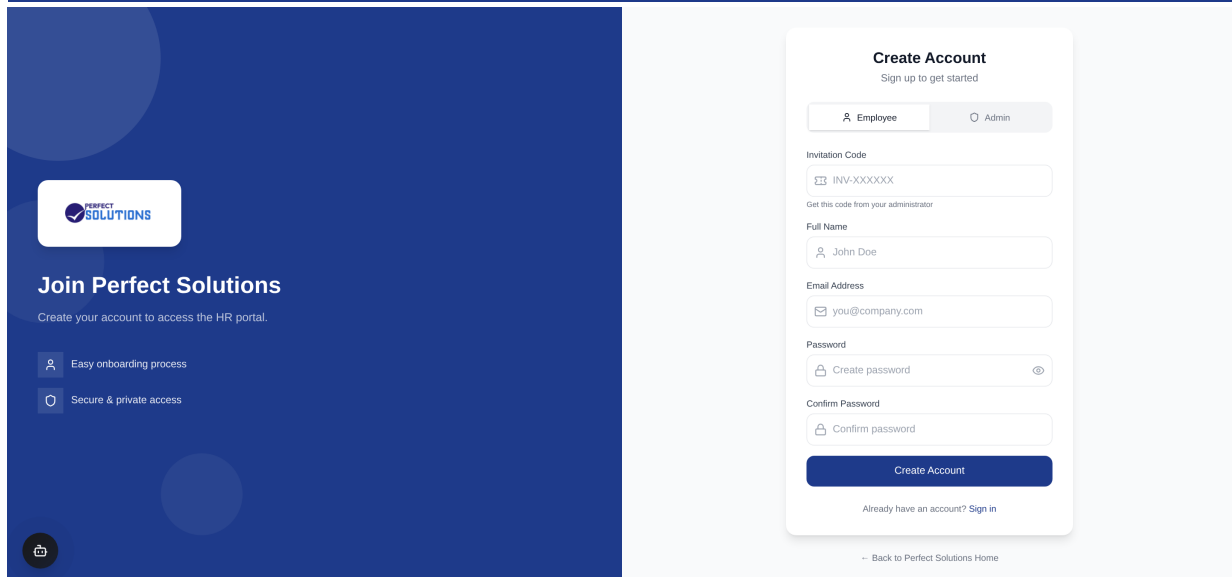
Sign in as Employee

Don't have an account? Sign up

Back to Perfect Solutions Home

Login experience inherits the tenant brand. Tenant slug captured implicitly; only Perfect Solutions credentials are accepted.

Screen A3 — Admin Signup (Invite-Code Gated)



Join Perfect Solutions

Create your account to access the HR portal.

- Easy onboarding process
- Secure & private access

Create Account
Sign up to get started

Employee Admin

Invitation Code
INV-XXXXXX
Get this code from your administrator

Full Name
John Doe

Email Address
you@company.com

Password
Create password

Confirm Password
Confirm password

Create Account

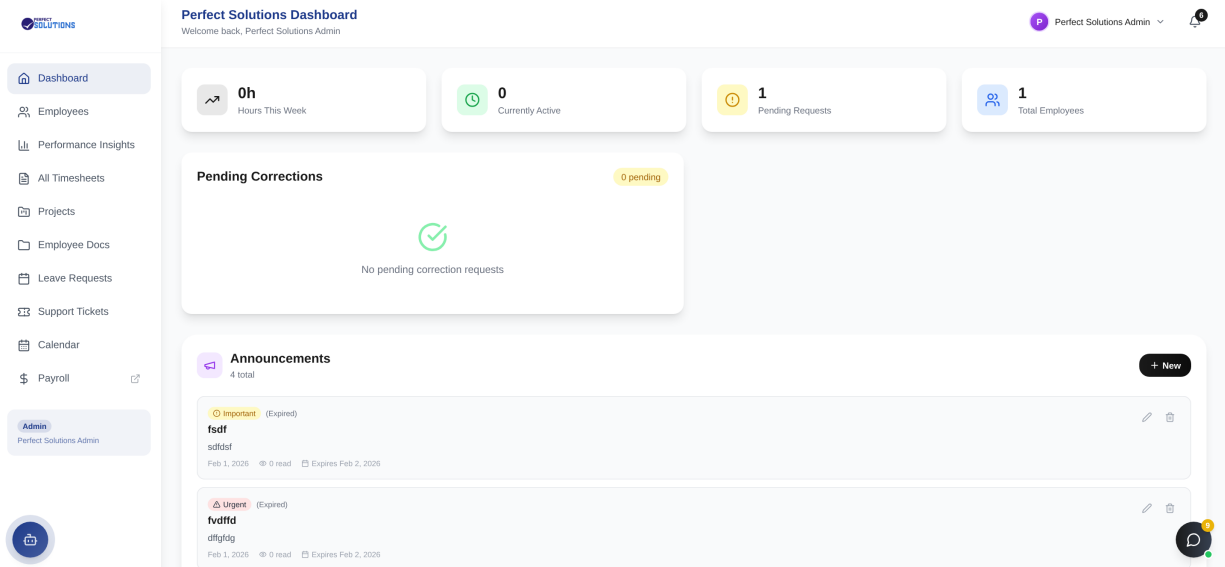
Already have an account? Sign in

Back to Perfect Solutions Home

New tenant admins must present a tenant-specific invite code to register, enforcing the secure onboarding flow.

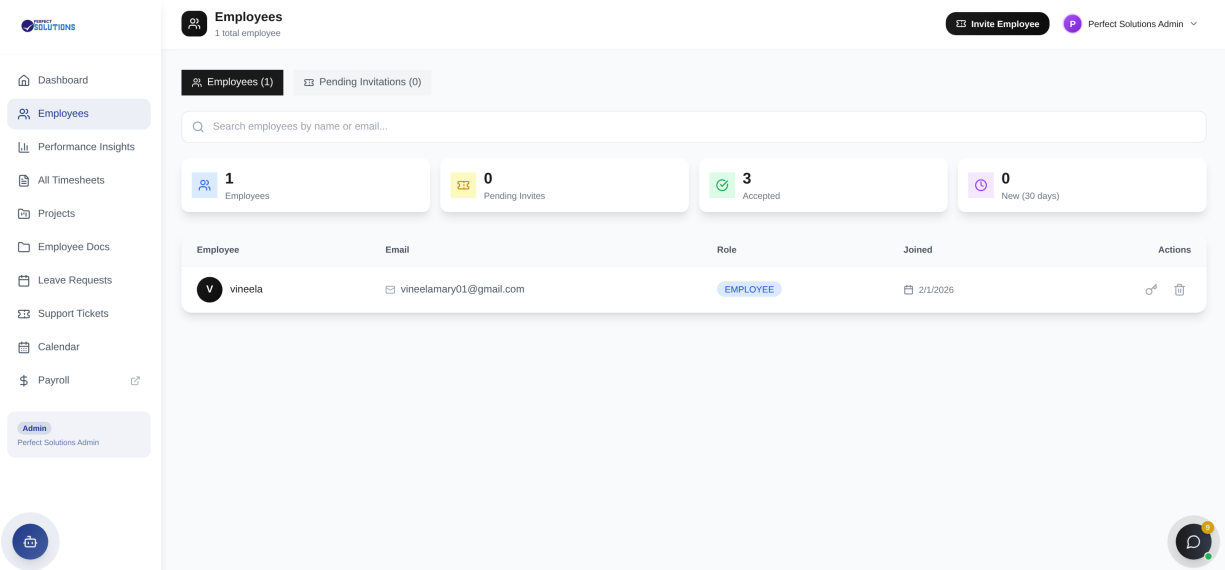
A.2 Admin / Manager Views

Screen A4 — Admin Dashboard



Unified admin landing — KPIs, recent activity, and quick access to operations modules.

Screen A5 — Employee Directory (Admin)



Searchable, filterable directory with role and department context. Drives every people-related workflow.

Screen A6 — Timesheet Approvals (Admin / Manager)

All Timesheets
Admin View

Audit Log Perfect Solutions Admin

Filters Employee: All Employees Start Date: mm/dd/yyyy End Date: mm/dd/yyyy Clear Export PDF Export CSV

No timesheet entries found

Pending timesheet review queue. Approve, reject, or request changes; bulk actions supported.

Screen A7 — Leave Approval Queue (Admin / Manager)

Leave Management
Review requests and manage leave settings

Requests 1 Settings

Pending 1 Approved Denied All Search by name or type... Export PDF

Perfect Solutions Admin VACATION Mar 1, 2026 - Mar 2, 2026 Reason: Test Pending Review Request

Pending leave requests with balance, overlap, and policy context for one-click decisions.

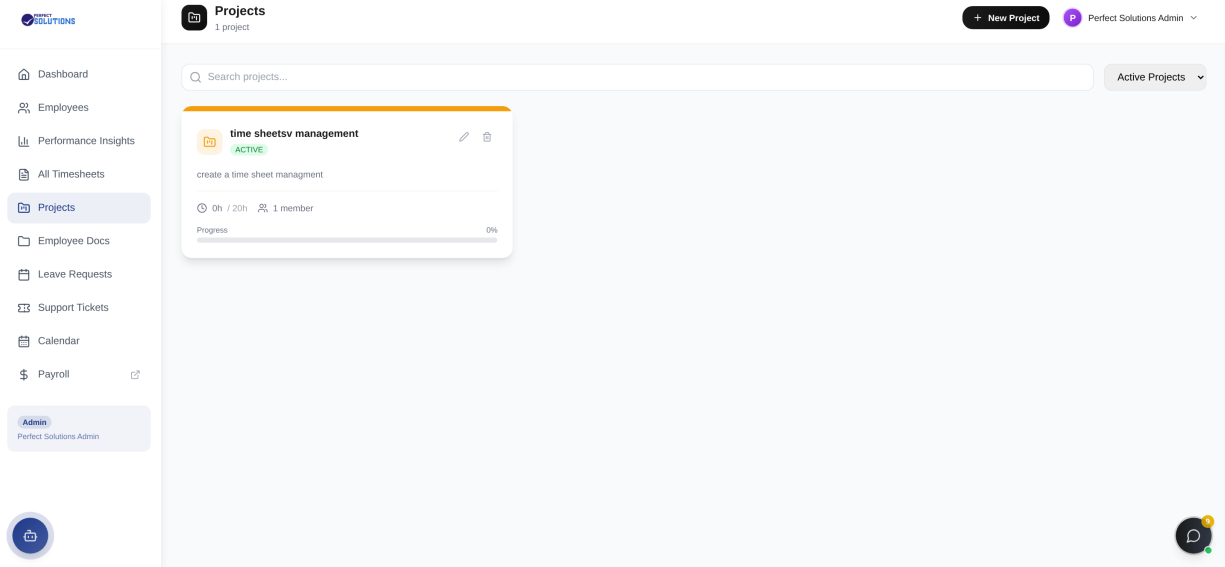
Screen A8 — Tickets / Helpdesk (Admin View)

Internal helpdesk with categorization, SLA visibility, and threaded responses.

Screen A9 — Unified Calendar

Approved leaves, company events, and scheduled meetings on a single calendar — Day, Week, Month, Agenda views.

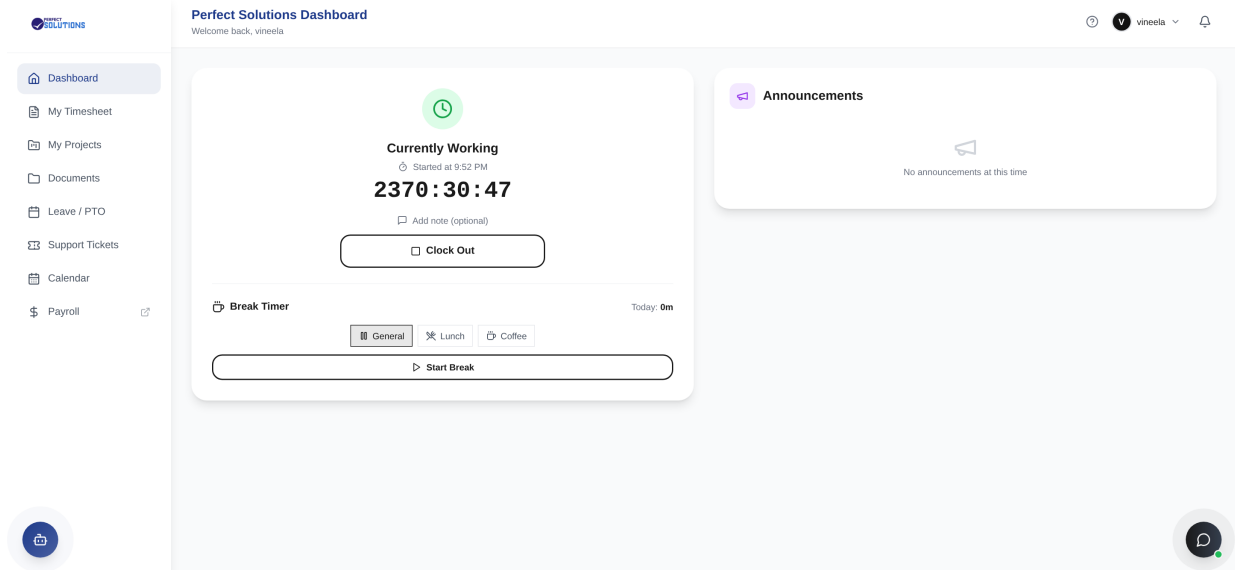
Screen A10 — Projects & Allocation



Project list with capacity and allocation overview — drives the utilization reports.

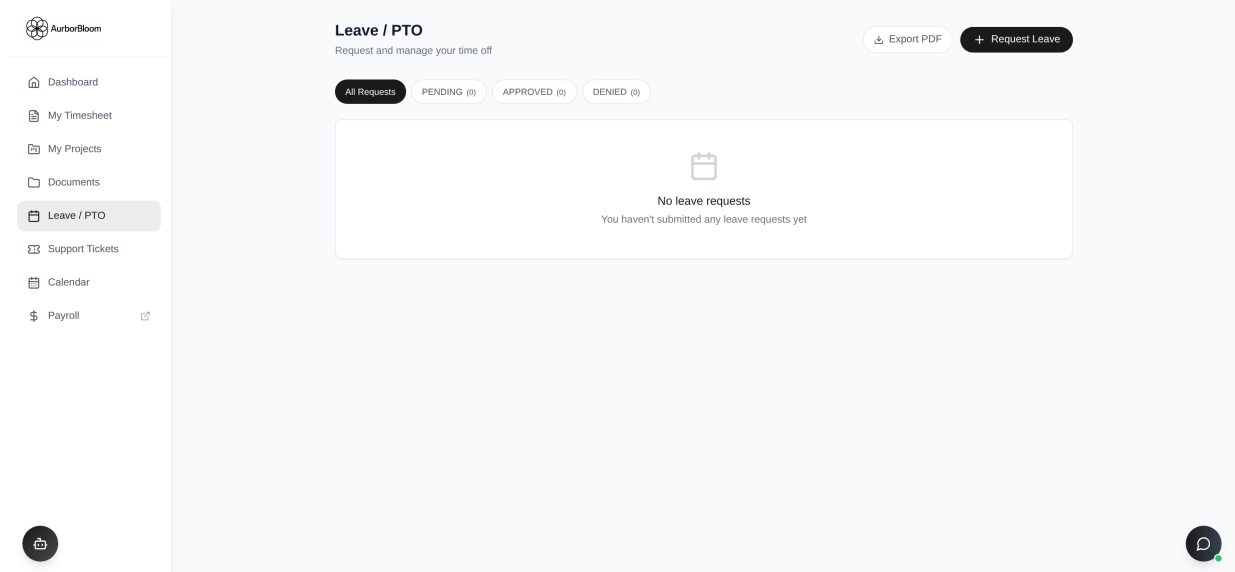
A.3 Employee Self-Service Views

Screen A11 — Employee Dashboard



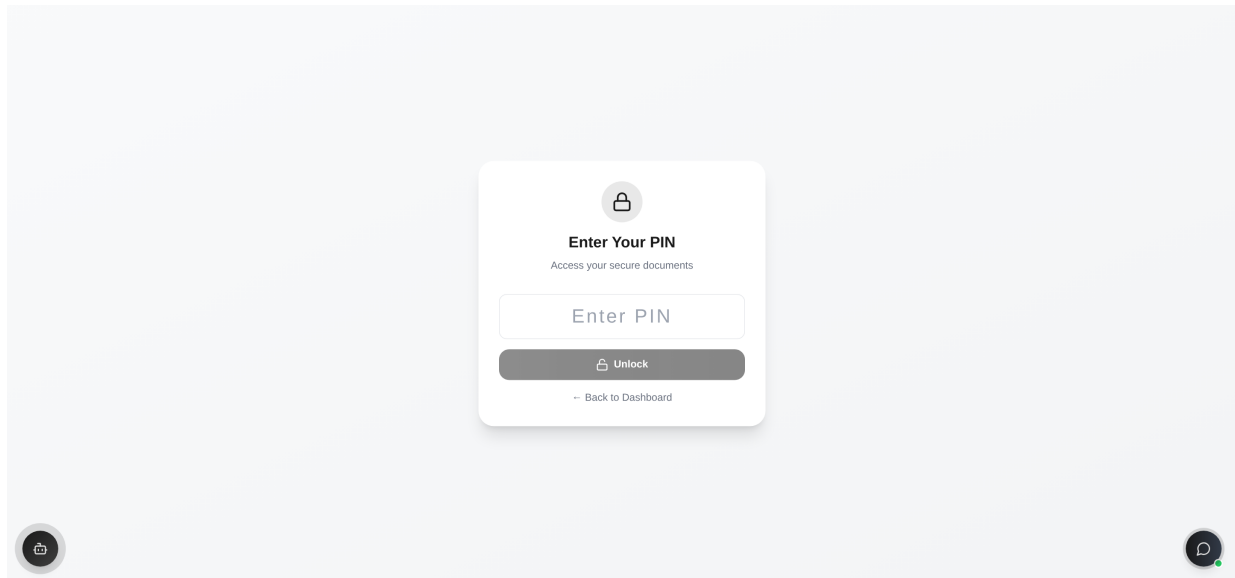
Self-service home: clock in/out, pending requests, announcements, quick actions.

Screen A12 — Employee Leave



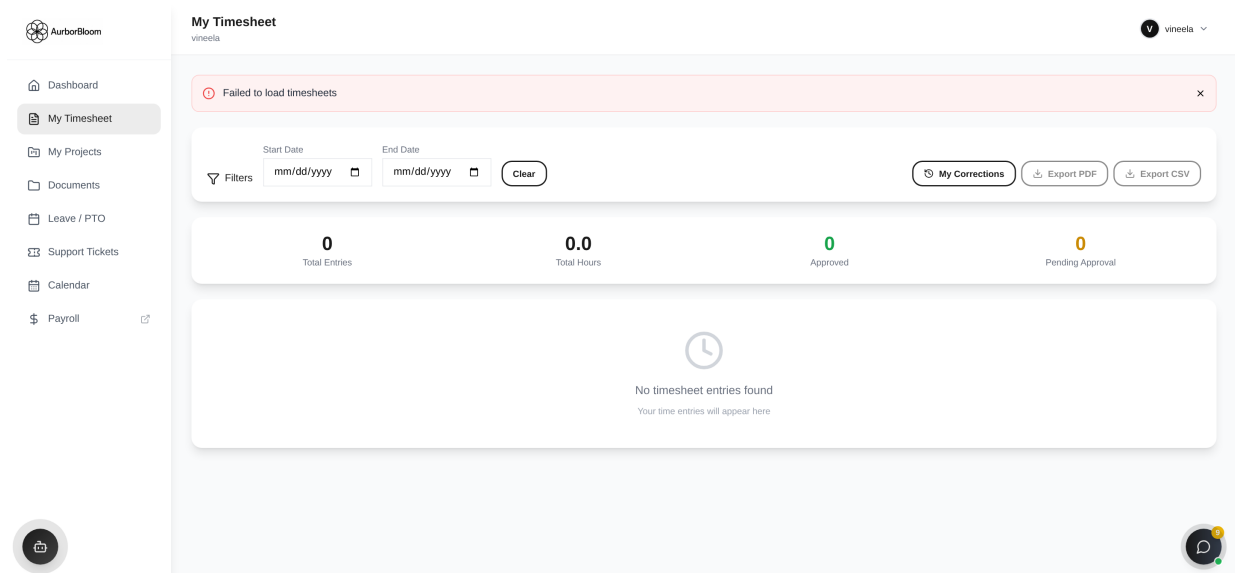
Real-time leave balance with request submission, status tracking, and PDF export.

Screen A13 — Document Vault



PIN-secured personal document vault — payslips, offer letters, signed policies.

Screen A14 — Employee Timesheet



Daily/weekly entry with project tagging, drafts, and submit-for-approval workflow.

Screen A15 — Employee Tickets

PERFECT SOLUTIONS

Dashboard

My Timesheet

My Projects

Documents

Leave / PTO

Support Tickets

Calendar

Payroll

Support Tickets

Get help from HR and IT support

Export PDF

+ New Ticket

vineela

Total Tickets

1

Active

0

Needs Response

0

Resolved

1

Q Search tickets...

Open0

No tickets

In Progress0

No tickets

Needs Your Response0

No tickets

Resolved1

TXT-20260201-0001

Regarding time sheets

Pending assignment

Feb 1

Raise and track internal support tickets; threaded comments and status updates.